

ChemComm

Chemical Communications

Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: J. H. Ha, B. Lee, M. Lee, T. Yim and S. H. Oh, *Chem. Commun.*, 2020, DOI: 10.1039/D0CC05611A.



This is an Accepted Manuscript, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this Accepted Manuscript with the edited and formatted Advance Article as soon as it is available.

You can find more information about Accepted Manuscripts in the [Information for Authors](#).

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard [Terms & Conditions](#) and the [Ethical guidelines](#) still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this Accepted Manuscript or any consequences arising from the use of any information it contains.

COMMUNICATION

Al-compatible boron-based electrolytes for rechargeable magnesium batteries

Jung Hoon Ha,^{a,1} Boeun Lee,^{a,1} Minseok Lee,^a Taeun Yim,^{*b} and Si Hyoung Oh^{*a,c}

Received 00th January 20xx
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

Galvanic couple-assisted dissolution of Mg metal in an ethereal solution containing tris(2H-hexafluoroisopropyl) borate was utilized to prepare an efficient Al-compatible electrolyte for rechargeable Mg batteries. The electrolyte exhibited conditioning-free Mg deposition, high oxidative stability (3.5 V vs. Mg/Mg²⁺), and a superb electrochemical performance with various cathodes on Al current collector.

Recently, rechargeable Mg batteries (RMBs) have emerged as an effective alternative to conventional Li-ion batteries (LIBs) used in advanced electric vehicles (EVs) and electricity storage systems (ESSs) for renewable energy power plants.^{1–5} Compared with LIBs, RMBs are cost-effective because they require earth-abundant Mg (2.33% availability) as a raw material. RMBs are also safer to use because of the intrinsic chemical stability of Mg metal in ambient environment and the dendrite-free nature of Mg deposition. Furthermore, Mg has a high volumetric capacity of 3,833 mA h cm^{–3}.^{6,7} Despite these conspicuous advantages, the advancement of the RMB technology has been limited. Particularly, slow development of an appropriate electrolyte that exhibits a reversible Mg plating–stripping behavior with a wide stability window and chemical compatibility toward cathode materials hinders the broad-scale application of RMBs. As the passive oxide layer on Mg metal serves as an ion-insulating layer, the water content in the electrolyte must be strictly controlled to obtain a reliable performance.^{1–5} To date, functional electrolytes are generally prepared through a careful combination of a strong base containing Mg ions (e.g., Grignard reagents, RMgX, R=alkyl or aryl, X=Cl, Br) with a Lewis acid such as

chloroalkylaluminum (e.g., R_xAlCl_{3–x}) in an ethereal solution, which forms various impromptu charge-transport species such as MgCl⁺ and [R_xAlCl_{4–x}][–].^{8–10} Several research groups have focused on the development of electrolytes comprising of all-inorganic-species to improve the oxidative stability of the electrolytes by controlling ligand activity.^{11–13} It was reported that these electrolytes were capable of efficient Mg plating–stripping and were stable up to ca. 3.0 V vs. Mg electrode. However, these electrolytes are highly corrosive because of the presence of chloride species, requiring corrosion-resistant current collectors, and their performance is critically affected even by small amount of water contamination.^{14–16}

Since early reports from Gregory *et al.* and Aurbach *et al.*^{9,17}, numerous efforts have been undertaken to develop efficient RMB electrolytes based on boron-centered anions. The chemistry of boron is fairly interesting; because boron behaves as an electron acceptor, it easily constitutes polyanionic compounds via coordination with electron-rich nucleophiles. For instance, Zhang *et al.* combined tri(3,5-dimethylphenyl)borane with PhMgCl in tetrahydrofuran (THF), resulting in a new electrolyte with excellent physicochemical properties.¹⁸ Mohtadi *et al.* utilized *closo*-borane clusters as a new class of electrolytes.¹⁹ Cui *et al.* reported a series of electrolytes based on boron-centered anions and applied them to Mg–S/Se batteries.^{20–22} These electrolytes were attractive, because boron-centered anions are considered to be more chemically stable than their aluminum counterparts, thus inducing fewer parasitic reactions. However, the presence of chloride ions could still result in unwanted corrosion of cell components.^{14–16} In this regard, RMB performance can be improved by using an electrolyte that exhibits corrosion resistance to metal current collectors. Herein, we prepare a chloride-free electrolyte through the catalytic dissolution of Mg metal containing boron compounds. This electrolyte exhibited efficient Mg plating, high oxidative stability of 3.5 V vs. Mg/Mg²⁺ in various current collectors, and compatibility with the Mo₆S₈ and S₈ cathodes.

^a Centre for Energy Storage Research, Korea Institute of Science and Technology, 5, Hwarang-ro 14-gil, Seongbuk-gu, Seoul, 02792, Korea. E-mail: sho74@kist.re.kr

^b Department of Chemistry, Incheon National University, 119, Academy-ro, Songdo-dong, Yeonsu-gu, Incheon, 22012, Korea. E-mail: yte0102@inu.ac.kr

^c Division of Energy & Environment Technology, Korea University of Science and Technology, Seoul, 02792, Korea

† Electronic Supplementary Information (ESI) available: See DOI: 10.1039/x0xx00000x

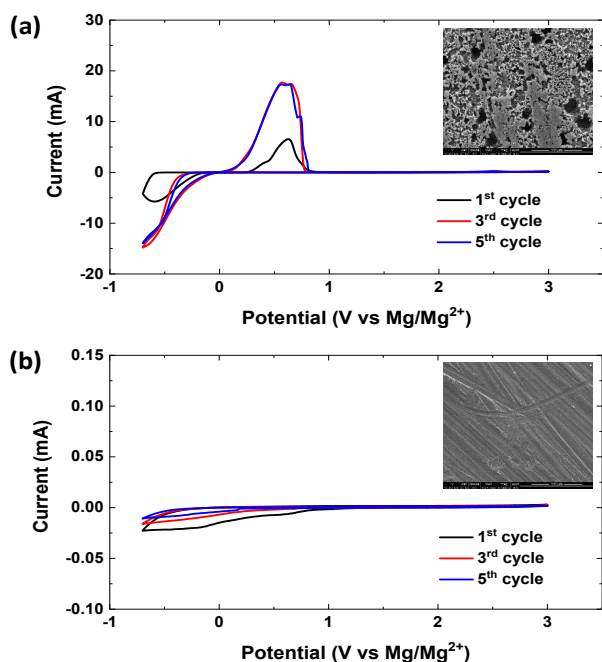


Fig. 1 Comparison of cyclic voltammograms for Mg electroplating-stripping in the electrolyte solution prepared (a) with CrCl_3 catalyst (electrolyte **A**) and (b) without the catalyst. The insets show the surface morphology of Mg plates used to prepare the electrolyte. Note that scale of y-axis is vastly different between (a) and (b).

The new electrolyte (electrolyte **A**) was prepared by dissolving Mg metal powder in 1,2-dimethoxyethane (DME) containing 0.3 M tris(2H-hexafluoroisopropyl) borate (THFPB), as described in the *Experimental details* in the ESI†. This was accomplished by the addition of a small amount of CrCl_3 catalyst in the DME solution, which formed Cr islet deposits on the Mg metal surface (Fig. S1, ESI†).^{13,23} These Cr-Mg galvanic couples significantly enhanced the Mg dissolution rate, as a Cr islet and the intimate Mg area acted as local cathodic and anodic centers, respectively, thereby increasing their exchange current densities. The ethereal solution containing THFPB served as an electrolyte.²³ As shown in Fig. 1 insets, rough-hewn surfaces with numerous corrosion pits are observed on a Mg metal plate dipped in the ethereal solution containing the CrCl_3 additive, while no apparent morphological variation was found on the one dipped in a solution without the catalyst. Cyclic voltammograms clearly show that reversible Mg plating-stripping was accomplished only in the electrolyte which was manufactured using CrCl_3 , with no noticeable electrochemical activity observed for the other case. This indicates that a significant concentration of Mg^{2+} ions in the DME solution can be achieved only with the addition of the catalyst.

The Mg electroplating-stripping behavior at a constant current of $15 \mu\text{A cm}^{-2}$ in electrolyte **A** with a Mo current collector was compared with those in two conventional high-performance electrolytes, *i.e.*, boron-centered-anion-based magnesium electrolyte (BCM)²⁰ and all-phenyl-complex

(APC).²⁴ BCM was prepared by combining THFPB and Mg5 in DME, while APC was synthesized by dissolving AlCl_3 in PhMgCl with THF. The results in Figs. 2a and 2b show that the stepwise voltage profiles could be divided into two separate regions. The initial voltage overshoot was associated with the nucleation overpotential (η_n), which was an additional energy barrier needed to form a new solid phase from the solution phase.²⁵ Compared to BCM, electrolyte **A** exhibited a considerably lower nucleation overpotential, particularly during the first cycle (214 vs. 305 mV). Furthermore, the associated coulombic efficiency of electrolyte **A** quickly stabilized within several cycles (92% vs. 86% at the 5th cycle). Similar tendency was also verified in the cyclic voltammetric measurement (Fig. S2, ESI†). This indicates that electrolyte **A** can be directly applied to practical cells once it is prepared without the need for a lengthy conditioning process.¹¹⁻¹³ The plateau region in the profiles represents an ohmic overpotential (η_o) for Mg plating-stripping in electrolyte **A** at a given current density, which was as low as that in APC (~40 and ~50 mV at the 5th cycle, respectively). The ionic conductivity of the electrolyte determined from the impedance spectra of Mg symmetrical cells was *ca.* 4.0 mS cm^{-1} , which was comparable to that of a 0.2 M APC solution (5.1 mS cm^{-1} , Fig. S3, ESI†). Linear sweep voltammetry (LSV) profiles in Fig. 2c indicates that the oxidative stability of electrolyte **A** with a Mo electrode was 3.3 V vs. Mg/Mg^{2+} , which was relatively higher than that of BCM and APC (2.6 and 2.8 V vs. Mg/Mg^{2+} , respectively). Furthermore, LSV performed with various metal current collectors as working electrodes (Fig. 2d and S4, ESI†) indicated that electrolyte **A** is considerably less corrosive than APC. In particular, aluminum, which is a common inexpensive current collector used in LIBs, was the most stable in electrolyte **A** (3.5 V vs. Mg/Mg^{2+}), but the least stable in APC (1.3 V vs. Mg/Mg^{2+}). This unusual stability can be attributed to the presence of boron-centered

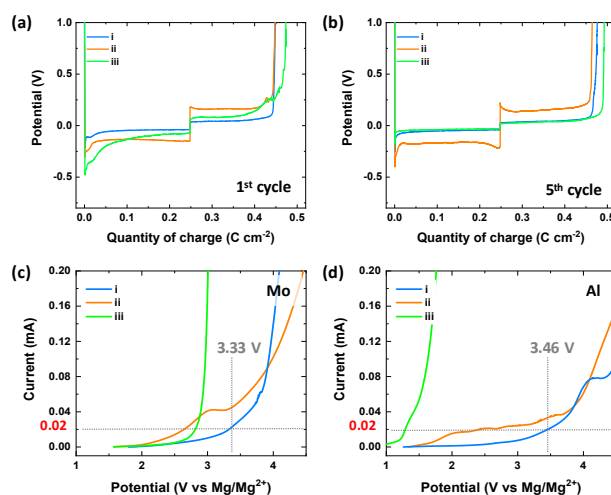


Fig. 2 Mg plating-stripping profiles at a constant current rate of $15 \mu\text{A cm}^{-2}$ in (i) electrolyte **A**, (ii) BCM, and (iii) APC solutions (a) at the first and 5th cycles. Linear sweep voltammograms for the various electrolytes in the evaluation of the oxidative stability with (c) Mo and (d) Al current collectors.

chloride-free anions in electrolyte **A**, making it less corrosive toward metal current collectors.¹⁴⁻¹⁶ Moreover, the applicability of electrolyte **A** to actual Mg batteries was evaluated by assembling coin-type cells, consisting of a Mo₆S₈ Chevrel phase²⁶⁻²⁸ or S₈@CMK-3 nano-composite²⁹ cathodes on commercial Al current collectors and Mg anode (Fig. 3). For Mo₆S₈, they exhibited typical plateau potentials owing to the intercalation of Mg²⁺ into the inner and outer ring vacancy sites in the structure, delivering a capacity of *ca.* 55 mA h g⁻¹ even at high current rate of 1C (128 mA g⁻¹) and a stable cycling performance for 300 cycles. S₈@CMK-3 cathode also showed a plateau potential from the conversion of S₈ to MgS and delivered *ca.* 500 mA h g⁻¹ at a current rate of C/10 (168 mA g⁻¹) in electrolyte **A**, which was much superior performance compared to other electrolytes (Fig. S5, ESI†). This confirms that electrolyte **A** is fully compatible with various cathode materials for Mg batteries and particularly, Al current collector.

The mechanism for the synthesis of electrolyte **A** was further analyzed by monitoring the variation in the concentration of Mg²⁺ ions in DME during electrolyte preparation. As shown in Fig. 4, once CrCl₃ was added, the concentration of Mg²⁺ ions began to increase gradually over a week. The enhanced corrosion of Mg metal was likely triggered by the formation of Cr-Mg galvanic couples.^{13,23} Cr³⁺ ions were reduced to form numerous Cr islands on the Mg surface (Fig. S1, ESI†), while the color of the solution changed from violet to transparent (Fig. 4b). The electrons were withdrawn from the Mg atoms adjacent to the Cr deposits, which became a local cathodic center. This led to a substantial increase in the exchange current density occurring on the Cr deposits, which involved a reduction of THFPB. The entire process caused an increase in the corrosion potential and Mg corrosion rate, as shown in Fig. S6, ESI†. It is noteworthy that the concentration of boron species in the electrolyte remained almost unchanged during electrolyte preparation. Therefore, the site responsible for the reduction reaction on the Cr surface was probably the hexafluoroisopropyl (HFIP) ligand, rather than the centered boron atom. Elemental analysis of the cross-section around the Cr island on the Mg metal was carried out through energy dispersive X-ray spectroscopy (EDS) with transmission electron microscopy (TEM, Fig. S1, ESI†). It

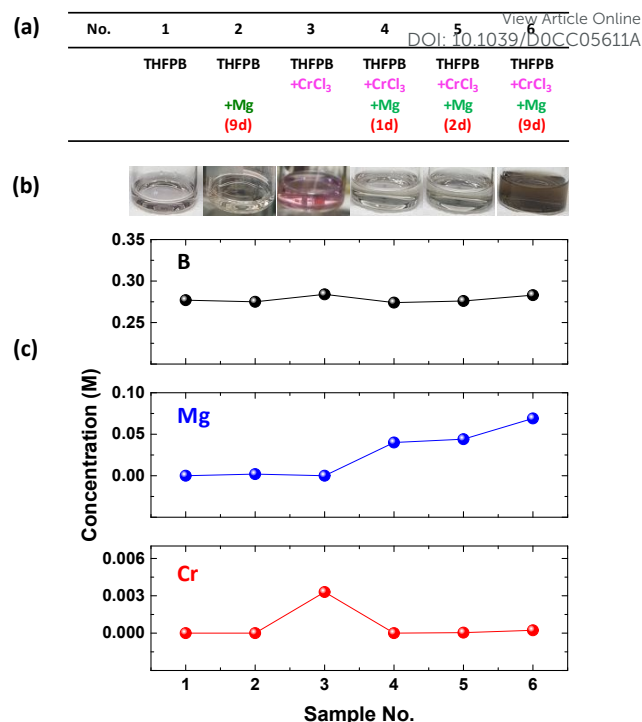


Fig. 4 (a) Sample descriptions for the various preparation steps for electrolyte **A** and (b) the corresponding color of the intermediate solutions. (c) Changes in the concentrations of B, Mg, and Cr in the solution at various preparation stages analyzed by ICP-OES. Samples were divided depending on the presence of the catalyst (CrCl₃) and Mg powder and the reaction duration after addition of Mg powder. Sample 2 was made from sample 1, while samples 4-6 were from sample 3 by adding Mg powder and reacting for desired time period.

shows that the Cr deposit was *ca.* 600 nm in diameter with a nonstoichiometric composition of CrO_x (*x* ~ 0.25). In addition, the presence of boron was not observed on the Cr island, supporting that it was the THFPB ligands that were likely reduced on the catalyst.

Nuclear Magnetic Resonance (NMR) studies were carried out to demonstrate the chemical role of THFPB and CrCl₃ during electrolyte synthesis (Fig. S7a, ESI†). In the ¹H NMR spectrum of THFPB in DME (Fig. S7a, ESI†), a ¹H peak at ~4.86 ppm was found, indicating that the ¹H position was attached to the THFPB ligand. The same peak was also observed in the ¹H NMR spectra of BCM (Fig. S7a, ESI†) which had THFPB as one of its components. Interestingly, the intrinsic ¹H peak completely disappeared in electrolyte **A**, implying that ¹H in the HFIP ligands chemically reacted with Mg metal by the CrCl₃ catalyst (Fig. S7a, ESI†). This could be rationalized through the proposed mechanism shown in Fig. S7b, ESI†. Once the acidic ¹H in the HFIP ligands was exposed to CrCl₃ with metallic Mg in the vicinity, CrCl₃ deprotonated HFIP ligands from THFPB. Further, the catalytic reaction with Mg resulted in MgH₂ formation as H could be easily removed from HFIP ligand

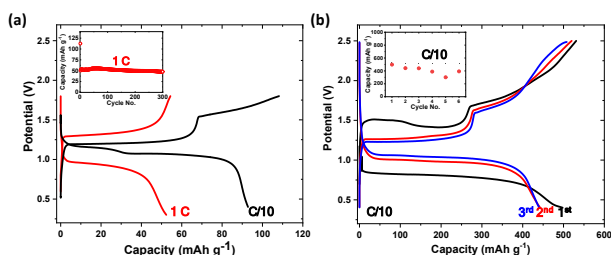


Fig. 3 (a) Discharge-charge profiles for the rechargeable Mg cells composed of Mo₆S₈ Chevrel phase on Al foil in the electrolyte **A**, when measured at a current rate of C/10 and 1C. Inset shows its cycling performance at 1C rate. (b) Discharge-charge profiles for Mg cells with CMK-3/S₈ cathode on Al foil in the electrolyte **A**, when measured at a current rate of C/10. Inset shows its cycling performance at C/10 rate.

because it is positioned at the most electrophilic reaction site. After the CrCl_3 -assisted reaction was completed, the resultant borate radical was converted to cationic borate (which is a highly unstable intermediate). A relatively nucleophilic DME molecule participated in the stabilization of the cationic borate through an intermolecular nucleophilic addition reaction, which resulted in the formation of a DME-added borate-oxonium second intermediate. Note that this oxonium intermediate is relatively unstable; hence, another DME participated in the elimination reaction, resulting in an OME-substituted borate complex (THFPB-OME). Note that this reaction continuously activated the Mg metal and formed products (THFPB-OME) are compatible with Mg batteries. These combinations composed of Mg, THFPB, and DME facilitated not only Mg activation but also stable cycling in Mg batteries.

In summary, a stable electrolyte for Mg batteries was prepared by a galvanic-couple-assisted Mg dissolution reaction in an ethereal solution containing THFPB. The electrolyte exhibited high reversibility, efficiency, and stability toward the Mo_6S_8 or S_8 cathode materials and various current collectors. This work provides an interesting strategy for preparing new viable electrolytes based on boron-functionalized compounds for RBMs.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgements

[†]J. Ha and B. Lee contributed equally to the work. This work was supported by the institutional program of KIST (2E30211), and by the National Research Foundation of Korea (NRF) (NRF-2019R1C1C1002249).

Notes and references

- H. D. Yoo, I. Shterenberg, Y. Gofer, G. Gershtinsky, N. Pour and D. Aurbach, *Energy Environ. Sci.*, 2013, **6**, 2265.
- R. Mohtadi, and F. Mizuno, *Beilstein J. Nanotechnol.*, 2014, **5**, 1291.
- Z. Zhao-Karger, and M. Fichtner, *MRS Commun.*, 2017, **7**, 770.
- M. Mao, T. Gao, S. Hou, and C. Wang, *Chem. Soc. Rev.*, 2018, **47**, 8804.
- P. Canepa, G. Sai Gautam, D. C. Hannah, R. Malik, M. Liu, K. G. Gallagher, K. A. Persson, G. Ceder, *Chem. Rev.*, 2017, **11**, 4287.
- H. D. Yoo, E. Markevich, G. Slitra, D. Sharon, and D. Aurbach, *Mater. Today*, 2014, **17**, 110.
- P. Saha, M. K. Datta, O. I. Velikokhatnyi, A. Manivannan, D. Alman, and P. N. Numta, *Prog. Mater. Sci.*, 2014, **66**, 1.
- D. Aurbach, Z. Lu, A. Schechter, Y. Gofer, H. Gizbar, R. Turgeman, Y. Cohen, M. Moshkovich, and E. Levi, *Nature*, 2000, **407**, 724.
- D. Aurbach, H. Gizbar, A. Schechter, O. Chusid, H. E. Gottlieb, Y. Gofer, and I. Goldberg, *J. Electrochem. Soc.*, 2002, **149**, A115.
- D. Aurbach, I. Weissman, Y. Gofer, and E. Levi, *Chem. Rec.*, 2003, **3**, 61.
- R. E. Doe, R. Han, J. Hwang, A. J. Gmitter, I. Shterenberg, H. D. Yoo, N. Pour, and D. Aurbach, *Chem. Commun.*, 2014, **50**, 243.
- C. J. Barile, E. C. Barile, K. R. Zavadil, R. G. Nuzzo, and A. A. Gewirth, *J. Phys. Chem. C*, 2014, **118**, 27623.
- J. H. Ha, B. Adams, J.-H. Cho, V. Duffort, J. H. Kim, K. Y. Chung, B. W. Cho, L. F. Nazar, and S. H. Oh, *J. Mater. Chem. A*, 2016, **4**, 7160.
- J. Muldoon, C. B. Bucur, A. G. Oliver, J. Zajicek, G. D. Allred, and W. C. Boggess, *Energy Environ. Sci.*, 2013, **6**, 482.
- J. Muldoon, C. B. Bucur, A. G. Oliver, T. Sugimoto, M. Matsui, H. S. Kim, G. D. Allred, J. Zajicek, and Y. Kotani, *Energy Environ. Sci.*, 2012, **5**, 5941.
- J. H. Ha, J.-H. Cho, J. H. Kim, B. W. Cho, and S. H. Oh, *J. Power Sources*, 2017, **355**, 90.
- T. D. Gregory, R. J. Hoffman, and R. C. Winterton, *J. Electrochem. Soc.*, 1990, **137**, 775.
- Y.-S. Guo, F. Zhang, J. Yang, F.-F. Wang, Y. NuLi, and S.-I. Hirano, *Energy Environ. Sci.*, 2012, **5**, 9100.
- T. J. Carter, R. Mohtadi, T. S. Arthur, F. Mizuno, R. Zhang, S. Shirai, and J. W. Kampf, *Angew. Chem. Int. Ed.*, 2014, **53**, 3173.
- Z. Zhang, Z. Cui, L. Qiao, J. Guan, H. Xu, X. Wang, P. Hu, H. Du, S. Li, X. Zhou, S. Dong, Z. Liu, G. Cui, and L. Chen, *Adv. Energy Mater.*, 2017, 1602055.
- A. Du, Z. Zhang, H. Qu, Z. Cui, L. Qiao, L. Wang, J. Chai, T. Lu, S. Dong, T. Dong, H. Xu, X. Zhou and G. Cui, *Energy Environ. Sci.*, 2017, **10**, 2616.
- H. Xu, Z. Zhang, Z. Cui, A. Du, C. Lu, S. Dong, J. Ma, X. Zhou, and G. Cui, *Electrochem. Commun.*, 2017, **83**, 72.
- J. H. Ha, J. Cho, J. H. Kim, B. W. Cho, H. C. Ham, and S. H. Oh, *J. Power Sources*, 2018, **398**, 120.
- O. Mizrahi, N. Amir, E. Pollak, O. Chusid, V. Marks, H. Gottlieb, L. Larush, E. Zinigrad, and D. Aurbach, *J. Electrochem. Soc.*, 2008, **155**, A103.
- H.-D. Lim, D. H. Kim, S. Park, M. E. Lee, H.-J. Jin, S. Yu, S. H. Oh, and Y. S. Yun, *ACS Appl. Mater. Interf.*, 2019, **11**, 38754.
- E. Gocke, R. Schöllhorn, G. Aselmann, and W. Müller-Warmuth, *Inorg. Chem.*, 1987, **26**, 1805.
- E. Gocke, W. Schramm, P. Dolscheid, and R. Schöllhorn, *J. Solid State Chem.*, 1987, **70**, 71.
- E. Levi, Y. Gofer, and D. Aurbach, *Chem. Mater.*, 2010, **22**, 860.
- X. Ji, K. T. Lee, and L. F. Nazar, *Nat. Mater.*, 2009, **8**, 500.

TOC abstract: Galvanic couple-assisted catalytic dissolution of Mg metal in dimethoxyethane containing tris(2H-hexafluoroisopropyl) borate lead to an efficient Al-compatible electrolyte for Mg batteries.

